# Data Visualisation

The Airbnb Market of Milan's Municipality 3

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## 1 The Question

For the Data Visualisation course, every group got one of Milan's nine municipalities (the *Municipi*) and had to tell the story of its Airbnb market using interactive visual analytics. We got **Municipality 3**, the north-east part of the city that goes from Porta Venezia to the city border. It is made of five NILs (*Nuclei di Identità Locale*, the official neighbourhood units used by the city): Buenos Aires-Venezia, Città Studi, Lambrate, Parco Forlanini-Ortica and Parco Lambro-Cimiano.

We organised the work around three simple questions:

1. How does Municipality 3 compare to the rest of Milan in terms of location quality, demand, price and revenue?
2. Inside the municipality, which NILs hold most of the listings, and which listings actually perform well?
3. What do hosts sell (in terms of accessibility and of the words they use) and does it change from one neighbourhood to another?

We built everything in **Tableau** and published the dashboards on Tableau Public, so they can be explored and filtered online. We also drew a network of the keywords used in the listing descriptions with **Gephi**.

This report explains what we did and how. Since a Tableau workbook is not easy to read without the software, we also rewrote all the calculated fields in Python and rebuilt every static view, so that anyone can check the numbers without a Tableau licence.

### *1.1 Where the data comes from*

The workbook combines six sources (Table 1). Some of them describe single listings, others describe whole NILs. The Inside Airbnb export is the main one, while the other five give us the territorial context and are linked to the listings through the NIL name or code.

**Table 1:** The six sources combined in the Tableau workbook. We recovered the last five from the data extracts packaged inside the workbook and saved them as CSV files in `data/raw/`.

| Source | Grain | Rows | Content |
|---|---|---|---|
| Inside Airbnb, Milan, Sept. 2025 | listing | 20,139 | price, room type, availability, reviews, review scores, estimated occupancy and revenue, keywords extracted from the descriptions |
| Comune di Milano, NIL boundaries | NIL | 88 | the shapes of the NILs, used for every map |
| Comune di Milano, `ds205` | NIL by year | 1,154 | resident population and demographics from 2011 to 2023 |
| Immobiliare.it | NIL | 86 | sale and rent prices per $m^2$, with their yearly trend |
| Driving distances | NIL | 88 | km and minutes by car to the Duomo, Linate and Malpensa |
| Road routes | route point | 6,932 | the GPS points of the routes from each NIL of Municipality 3 to the three destinations |

### *1.2 What we had to fix in the data*

Before starting, we found four problems in the data.

- **Currency.** The price column is called "price in dollars", but Inside Airbnb uses the local currency, so all the prices in this report are in euro.
- **Extreme prices.** A few listings ask for unrealistic prices: 4.3% of them cost more than €500 per night, 129 cost more than €2,000 and the most expensive one asks €65,500. For this reason the Tableau metrics leave out listings at €2,000 or more when computing average prices, the price box plot only shows listings up to €500, and we always report the median next to the mean.
- **A broken NIL name.** One NIL name has an encoding error in the export (`CITTÃ€` instead of `CITTA`), and seven listings end up without a municipality until we fix it.
- **Glued keywords.** In the keyword column some words are stuck to the ending "ight" (for example `metersight` or `perfectight`). These words appeared as separate nodes in the original Gephi graph, so we fixed them before rebuilding the network (Section 6).

## 2 How We Built It

### *2.1 From the Tableau workbook to Python*

The Tableau workbook (`tableau/Data_Viz_Project_2.twbx`) has six worksheets and one dashboard. Unfortunately the Gephi project was not saved, and one of our views (the revenue treemap) is not inside the workbook. To make the project complete and easy to check, we rebuilt the analysis with three Python scripts that only use the raw files saved in the repository:

- `prepare_data.py` cleans the listings, rewrites the Tableau calculated fields and saves the tables by listing, NIL and municipality in `data/processed/`;
- `make_figures.py` redraws the Tableau views as static figures (Figures 1 to 7);
- `build_text_network.py` rebuilds the Gephi keyword network (Figure 8).

The Python figures give exactly the same numbers as Tableau. For example, the average location score is 4.87 for Municipality 1 and 4.76 for Municipality 3 both in the workbook and in the script.

### *2.2 The calculated fields*

The workbook relies on a few calculated fields (Table 2). The most important one is the metric switch of the dashboard. A single parameter (*Select Metric*) controls one calculated field (*Selected Metric*), which sets both the colour of the map and the length of the bars. A second parameter (*Drill Municipality*) changes the bar chart from a ranking of the municipalities into a ranking of the NILs inside the chosen municipality. This is how the dashboard moves from the whole city to Municipality 3 without needing a second sheet.

**Table 2:** The calculated fields of the Tableau workbook, rewritten in `scripts/prepare_data.py`.

| Field | Definition |
|---|---|
| NIL Municipality | assigns each of the 88 NILs to one of the nine municipalities |
| Selected Metric | switches between average location score, average reviews per month, average price (listings under €2,000) and average revenue proxy, based on the *Select Metric* parameter |
| Revenue proxy | price × reviews per month × 12, for listings under €2,000 with at least one review per month |
| Drill Dimension | shows the municipalities when *Drill Municipality* is set to All, otherwise the NILs of the chosen municipality |
| Performance Index | rating × reviews in the last 12 months / price, set to 0 when it cannot be computed |
| Performance Tier | *New / Unproven* if the index is below 0.05, *Solid* if below 0.5, *Star Performer* otherwise |
| Price Tier | under €100, €100 to 150, €150 to 200, over €200 |

The Performance Index gives a higher score to listings that are well rated, booked often (we use recent reviews as a proxy for bookings) and cheap compared to that demand. In other words, it measures how much value a listing gives for each euro, not how much money it makes. With its thresholds, Milan's listings split into three groups of similar size (37% New / Unproven, 33% Solid, 30% Star Performer). Listings with no reviews in the last twelve months get a score of zero, so they automatically fall into the first group.

## 3 Municipality 3 Compared to the Rest of Milan

The main Tableau dashboard (*Territorial Context and Analysis*) is the centre of our work, and it is still online on Tableau Public. The map colours each area based on the selected metric, while the bar chart below ranks the areas on the same metric. Clicking on a municipality in one of the two views filters the other one.

Municipality 3 has 2,260 listings, which is 11.2% of all the listings in Milan and the fourth largest number among the nine municipalities. On all four metrics of the dashboard it always comes second (Figures 1 and 2). It is second only to the historic centre (Municipality 1) for location score (4.76 against 4.87), average price per night (€165 against €268) and revenue proxy (€3,568 against €4,942 per year), and second only to Municipality 2 for reviews per month (1.98 against 2.23).

Put simply, Municipality 3 is the closest alternative to the centre. Guests rate its locations almost as well as the centre, they book it more often, and they pay about 40% less per night. The share of entire homes (89%) is in line with the rest of the city.

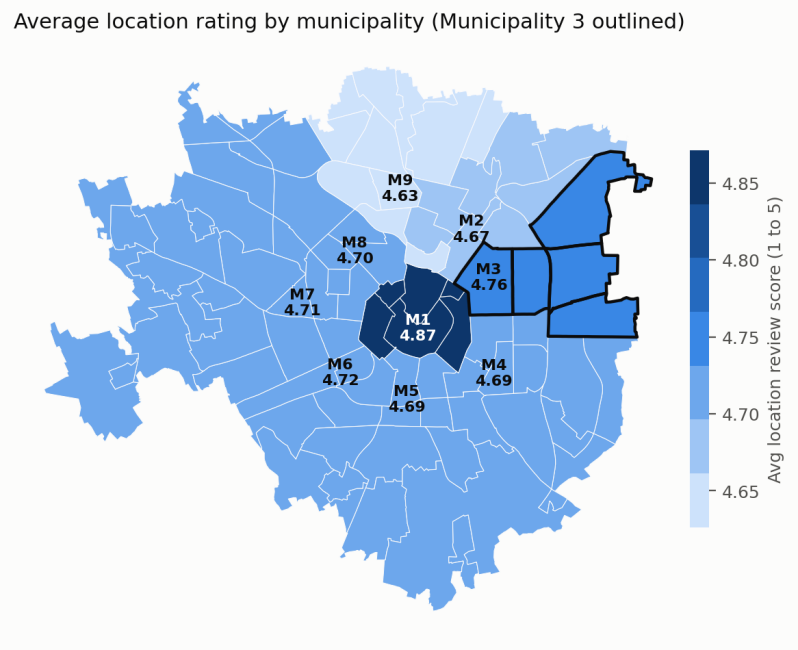



**Figure 1:** Average location score by municipality, with Municipality 3 outlined.

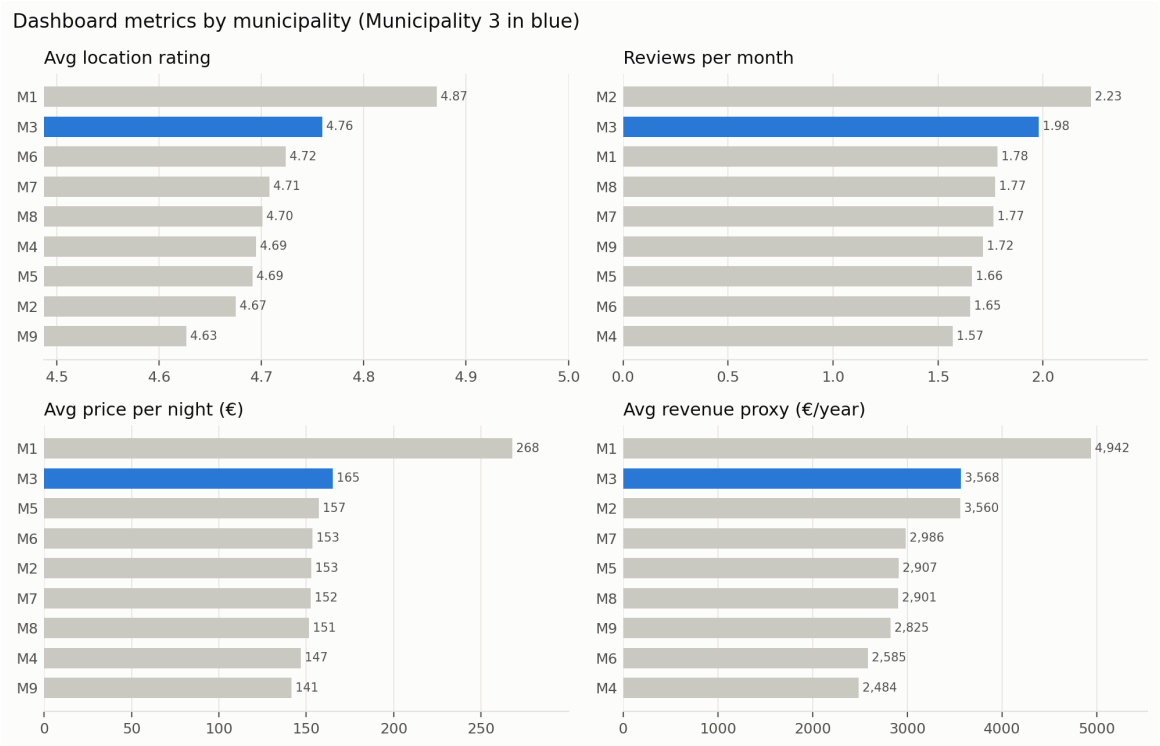



**Figure 2:** The four metrics of the dashboard for every municipality. The axis of the location score starts at 4.5, otherwise the differences on a 1 to 5 scale would not be visible.

## 4 Inside Municipality 3

### *4.1 Where the listings are*

The listings of the municipality are very concentrated. Buenos Aires-Venezia alone has 72% of them (1,633), Città Studi follows with 18%, and the other three NILs share the remaining 10%. Figure 3 compares this supply with the residential context. Buenos Aires-Venezia has 25.9 listings every 1,000 residents, almost twice the city average of 14.2, and it also has the highest sale price of the municipality (€7,902 per $m^2$, 12th out of all the NILs of Milan). Moving from Porta Venezia towards the city border, both the number of short-term rentals and the value of houses go down.

The resident population of the three largest NILs has stayed stable between 2011 and 2023. The big percentage increase of Parco Forlanini-Ortica is due to its very small size: it went from about 1,000 residents in 2011 to 2,454 in 2023, with a jump in 2017.

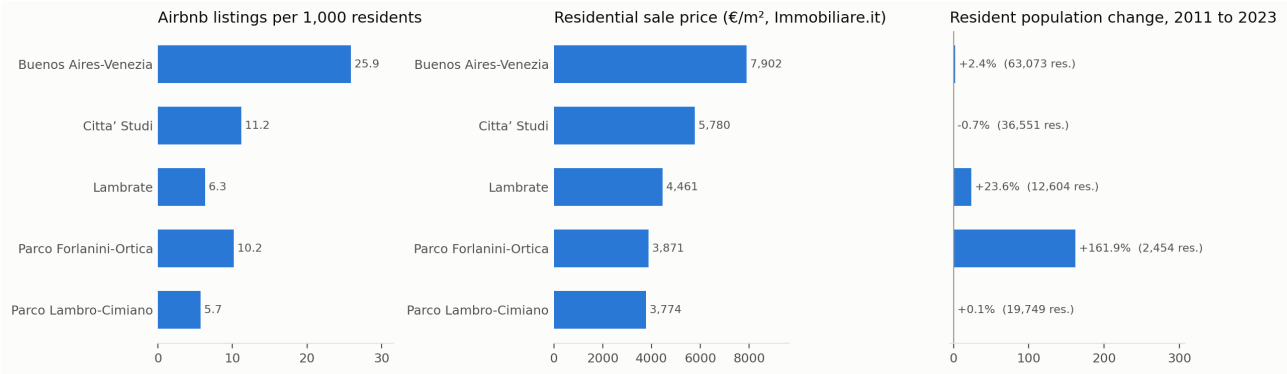



**Figure 3:** The NILs of Municipality 3: number of Airbnb listings per resident, sale price of houses (Immobiliare.it) and change in resident population.

### *4.2 Which listings perform*

The *Matrix + Dots* view in Tableau puts every listing in a grid of NILs and Performance Tiers, and sets the size and the colour of each dot based on its Performance Index. Figure 4 shows the same grid as percentages.

The mix of tiers is quite similar across the NILs. Star Performers are one third of the listings in the two largest NILs (33% both in Buenos Aires-Venezia and in Città Studi) and between 20% and 28% in the three smaller ones, while New / Unproven listings stay between 34% and 42% everywhere. This means that performance does not explain why the listings are so concentrated: Buenos Aires-Venezia does not have better listings than Città Studi, it simply has four times as many. Compared to the rest of the city, Municipality 3 has more Star Performers (32%) than the centre (22%), where high prices push the index down.

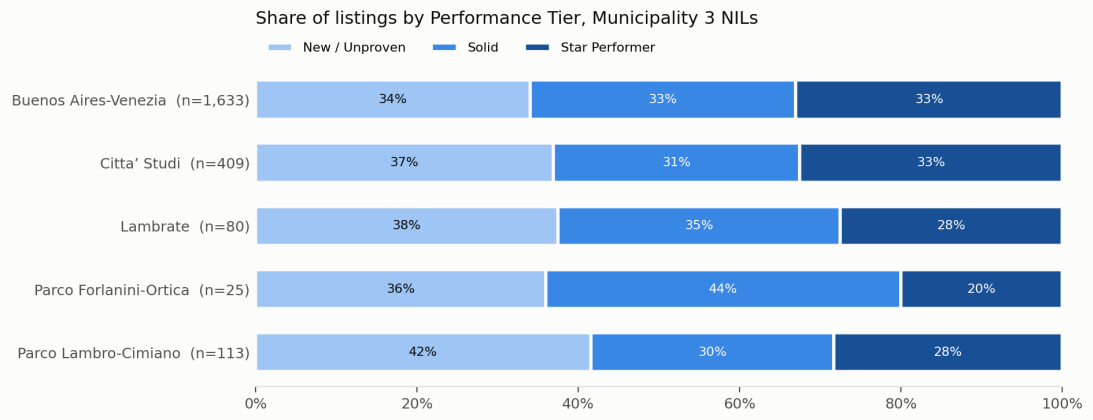



**Figure 4:** Share of listings in each Performance Tier in the five NILs of Municipality 3.

### *4.3 Price and revenue in the three largest NILs*

For the deeper analysis we focused on the three NILs with the most listings: Buenos Aires-Venezia, Città Studi and Parco Lambro-Cimiano. Prices follow the same pattern as house values (Figure 5). The median price per night is €138 in Buenos Aires-Venezia, higher than the Milan median of €125, while it goes down to €104 in Città Studi and €91 in Parco Lambro-Cimiano. Buenos Aires-Venezia also has the widest spread of prices, which makes sense for a market that mixes standard apartments with a more expensive segment around Porta Venezia and Corso Buenos Aires.

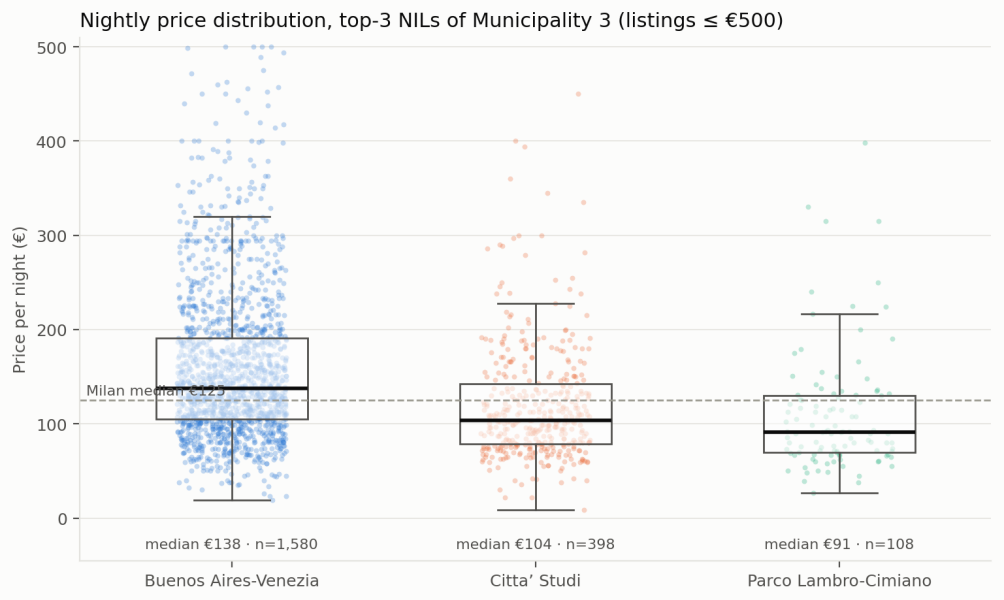



**Figure 5:** Distribution of the price per night in the three largest NILs (listings up to €500, one dot for each listing, whiskers at 1.5 times the interquartile range).

The revenue treemap (Figure 6) uses the revenue that Inside Airbnb estimates for the last 365 days. Entire rental units make most of the money everywhere: €15.9M out of the €21.4M produced by the three main property types in Buenos Aires-Venezia, and €5.2M out of €6.0M in Città Studi. Private rooms are among the three main types only in Città Studi, which is a university area where renting single rooms is a real niche.

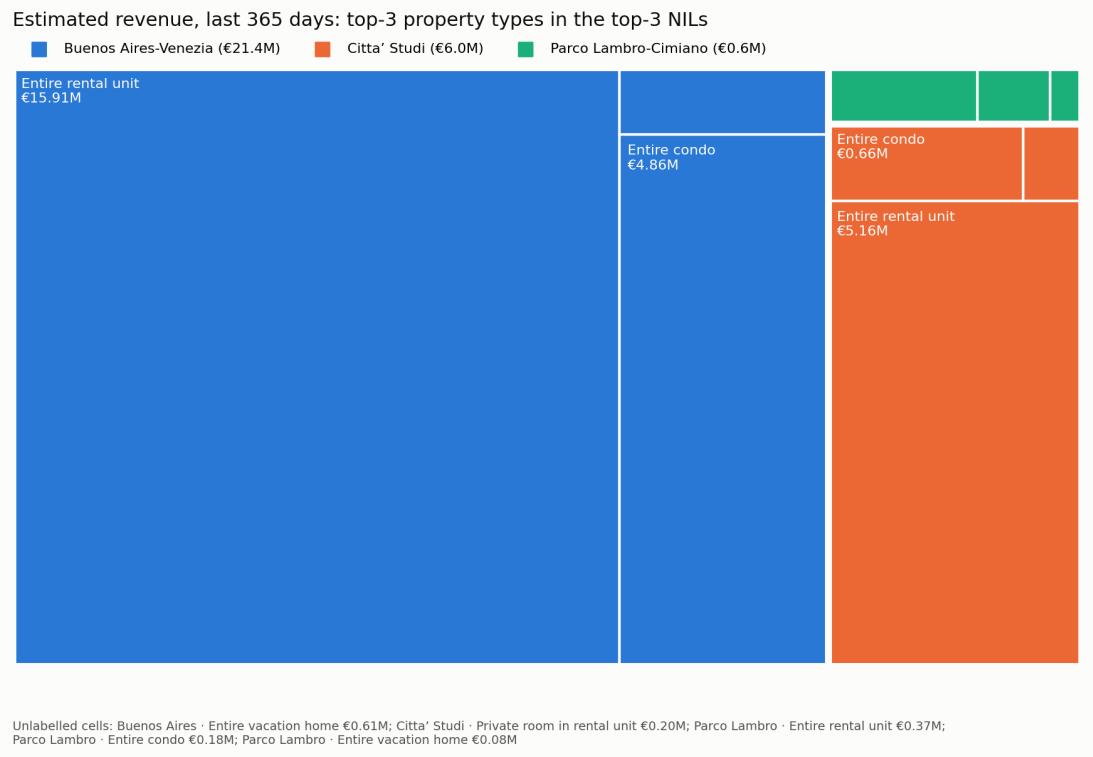



**Figure 6:** Estimated yearly revenue of the three main property types in each of the three largest NILs.

## 5 How Easy It Is to Get Around

The *Connection Map* sheet draws the real road route from a chosen NIL to the city centre and to the two airports, and a parameter lets the user change the starting NIL. Figure 7 shows the routes from Città Studi and compares the driving times of all five NILs.

Being in the east of the city is an advantage: every NIL is between 9 and 20 minutes from Linate, and all of them except Buenos Aires-Venezia are closer to Linate than to the Duomo. Buenos Aires-Venezia is the only

NIL within 15 minutes of the centre, which fits with its higher prices and its larger number of listings. Malpensa is about 50 minutes away from every NIL, so it does not make any difference between them.

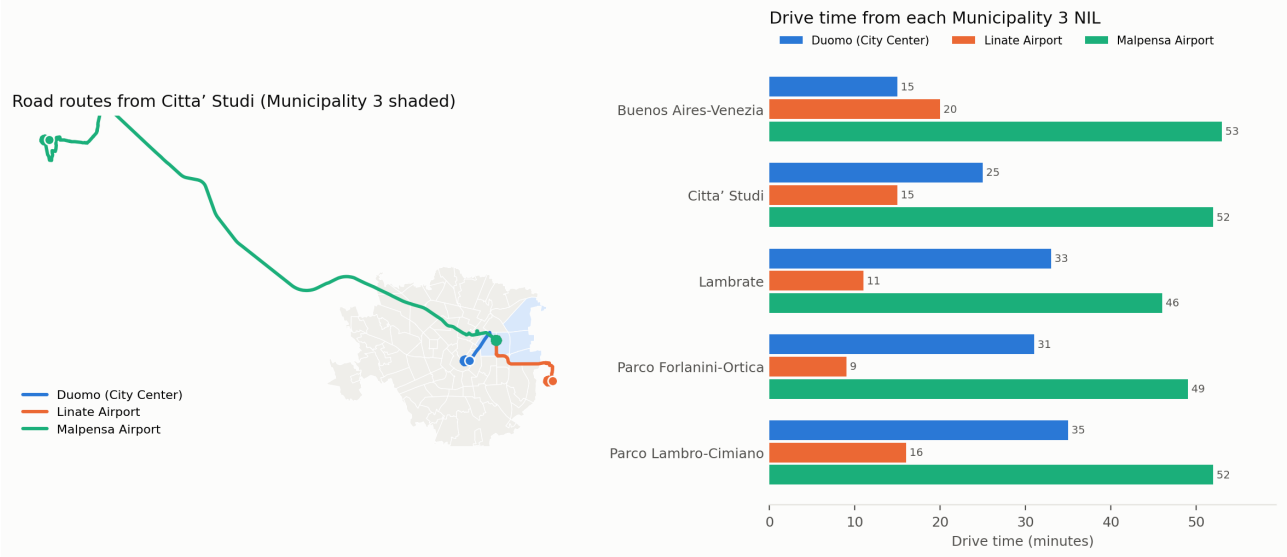



**Figure 7:** Road routes from Città Studi to the three destinations, and driving times from each NIL of Municipality 3.

## 6 What Hosts Say: the Keyword Network

In the last part of the analysis we wanted to see if hosts in different NILs describe their listings in different ways. The original graph was made in Gephi using the keywords extracted from the listing descriptions, and Figure 8 is our rebuilt version in Python.

The graph connects three NILs (Buenos Aires-Venezia, Città Studi and Parco Forlanini-Ortica, the same ones as in the original) to every keyword used by at least 8% of their listings. The thicker the line, the larger the share of listings using that keyword. After fixing the glued "ight" words and removing ten generic words (like *milan*, *apartment* or *room*), the graph has 33 nodes and 66 links. To find groups of keywords we used Louvain community detection (the same method behind the *Modularity Class* in Gephi) with a fixed seed, and it finds three communities (modularity 0.20), one around each NIL.

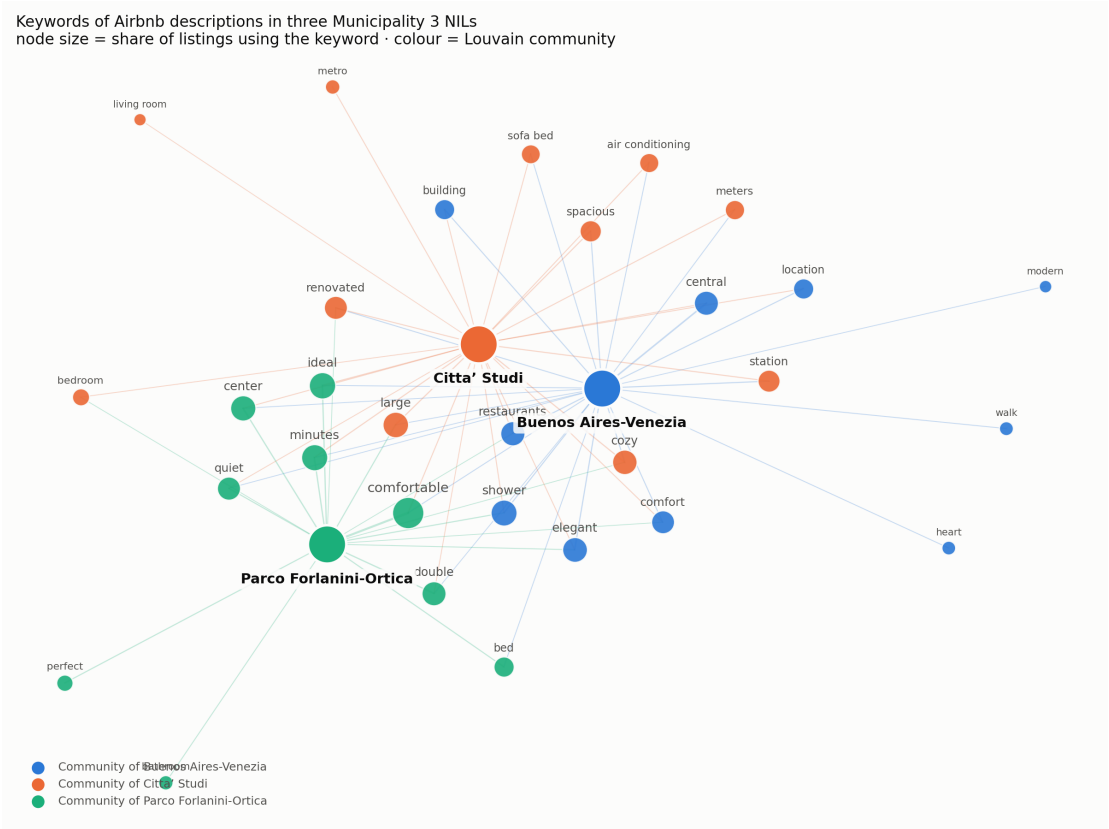



**Figure 8:** Keyword network of the listing descriptions, rebuilt from the `tags` column. The size of a node shows how often a keyword is used, the colour shows its community.

The words used by hosts follow the same lines as the rest of the analysis:

- in **Buenos Aires-Venezia** hosts sell location and atmosphere (*central*, *heart*, *walk*, *restaurants*, *elegant*, *modern*);
- in **Città Studi** they sell space and practicality (*large*, *spacious*, *renovated*, *sofa bed*, *metro*, *station*);
- in **Parco Forlanini-Ortica**, where listings are few and far from the centre, they sell comfort and quiet (*comfortable*, *quiet*, *minutes* from the centre, *double* bed).

Still, 23 of the 30 keywords are used in at least two NILs, so the three groups differ more in what they stress than in the words they use. Parco Forlanini-Ortica has only 25 listings, so its percentages are less reliable than those of the other two NILs.

## 7 How We Designed the Views

A few ideas guided our design choices, both in Tableau and in the Python version.

- **One view, many metrics.** Instead of making four different maps, the dashboard uses a parameter to switch metric. The map stays the same while the question changes, so comparing metrics is immediate.
- **Zooming in instead of duplicating.** The same bar chart ranks either the municipalities or the NILs depending on the drill parameter, so the audience stays on one dashboard while the story moves closer.
- **Interactive views online.** Publishing the dashboards on Tableau Public allows anyone to filter the data and explore it by themselves, without installing any software.
- **Colours with a clear job.** Quantities use a single colour going from light to dark, categories (NILs, destinations, communities) use a fixed set of colours that people with colour vision deficiencies can tell apart, and in the ranking charts Municipality 3 is the only coloured bar.
- **Showing the whole distribution.** We show prices as dots and box plots instead of simple averages, because a few extreme prices would make an average misleading.

## 8 What the Analysis Cannot Tell Us

This analysis describes the market at a single point in time, since it is based on one Inside Airbnb snapshot. It cannot show seasonal patterns or how the market changes over the years. Estimated revenue and occupancy

come from Inside Airbnb's own models, and our revenue proxy (price × reviews per month × 12) assumes that every review corresponds to one booked night. For this reason both numbers are useful to rank areas, not to measure real earnings.

The Performance Index rewards cheap listings by design and gives no credit to new listings or to listings that guests rarely review. The keyword network depends on a keyword extraction done before our work, whose exact method is not documented: our Python version rebuilds the structure of the Gephi graph but not its exact layout. Finally, the population data uses the 2011 to 2023 NIL series of the Comune di Milano, where some names are slightly different from the current list of NILs, so we linked the two using the official NIL code.

## Wrapping Up

Municipality 3 is the second short-term rental market in Milan. It comes right after the historic centre for location score, price and revenue, and right after Municipality 2 for demand, and it offers guests an almost central stay for a much lower price. Most of its listings are in Buenos Aires-Venezia, which is the NIL closest to the Duomo, with the most expensive houses and almost twice as many listings per resident as the city average. The quality of the listings, however, is spread fairly evenly across the municipality. Moving east, Città Studi and the other NILs compete on space, price and closeness to Linate rather than on being central, and the words their hosts use reflect this.